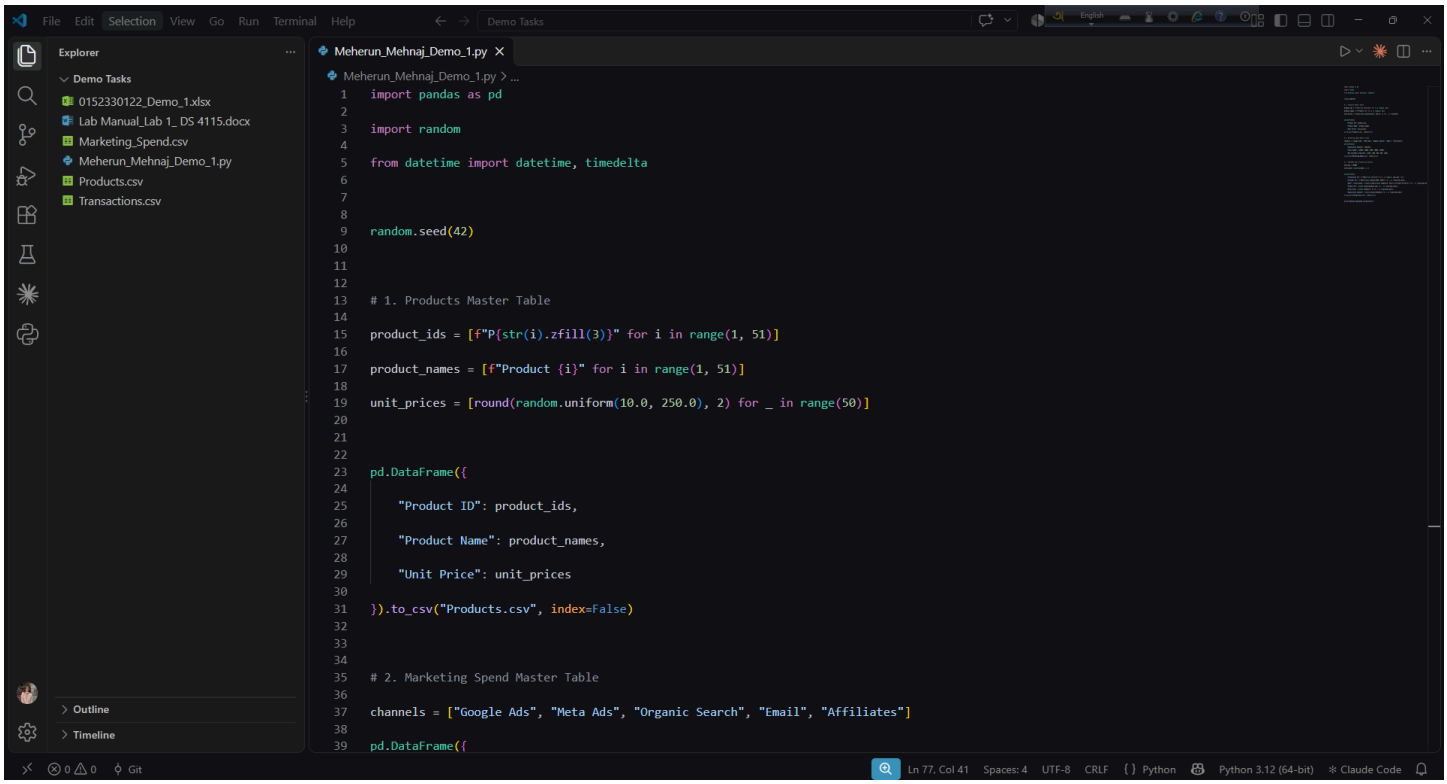
Step 1: Creating CSV files and codes




```python
import pandas as pd

import random

from datetime import datetime, timedelta



random.seed(42)



# 1. Products Master Table

product_ids = [f"P{str(i).zfill(3)}" for i in range(1, 51)]

product_names = [f"Product {i}" for i in range(1, 51)]

unit_prices = [round(random.uniform(10.0, 250.0), 2) for _ in range(50)]



pd.DataFrame({

    "Product ID": product_ids,

    "Product Name": product_names,

    "Unit Price": unit_prices

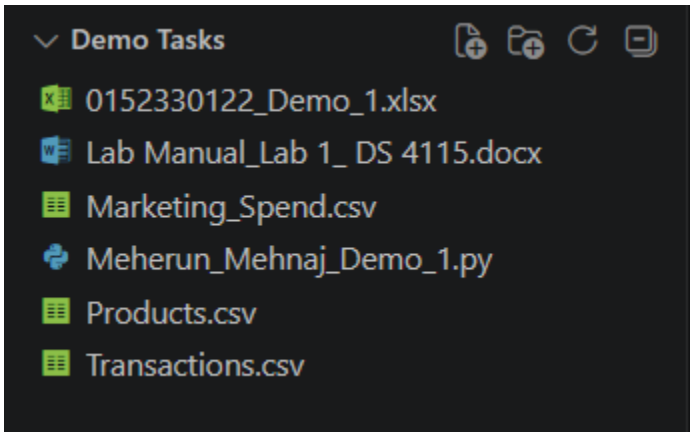
}).to_csv("Products.csv", index=False)



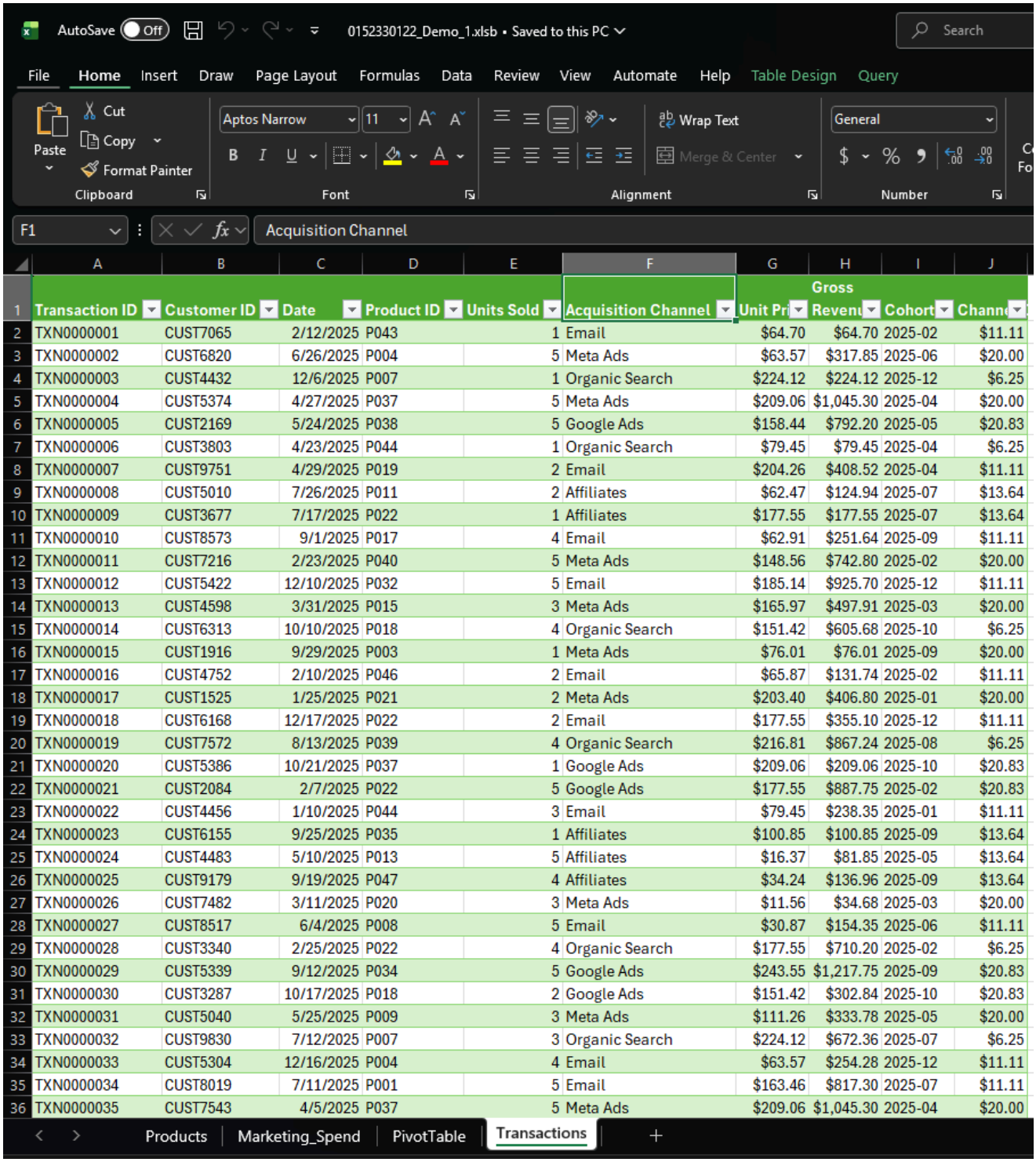
# 2. Marketing Spend Master Table

channels = ["Google Ads", "Meta Ads", "Organic Search", "Email", "Affiliates"]

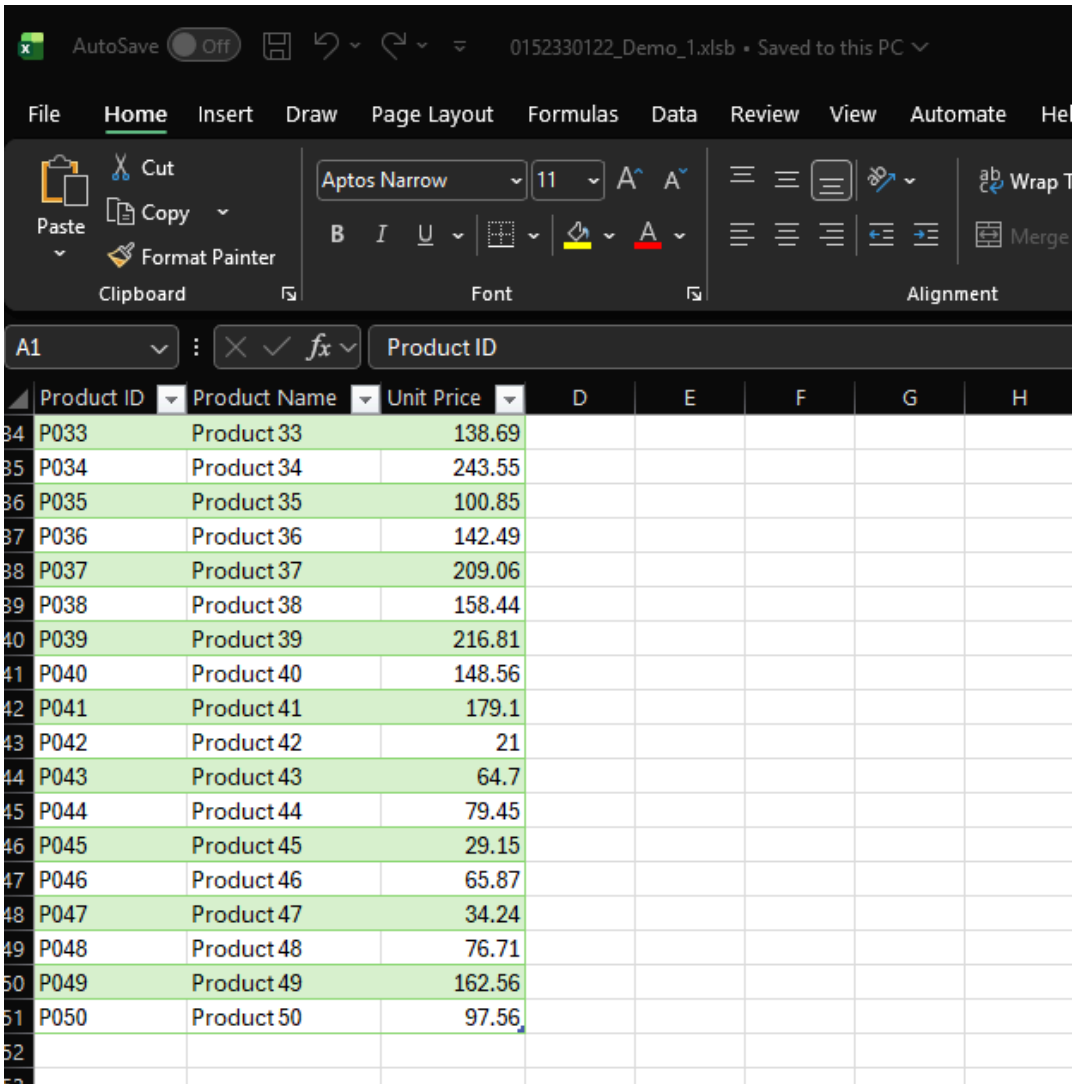
pd.DataFrame({
```



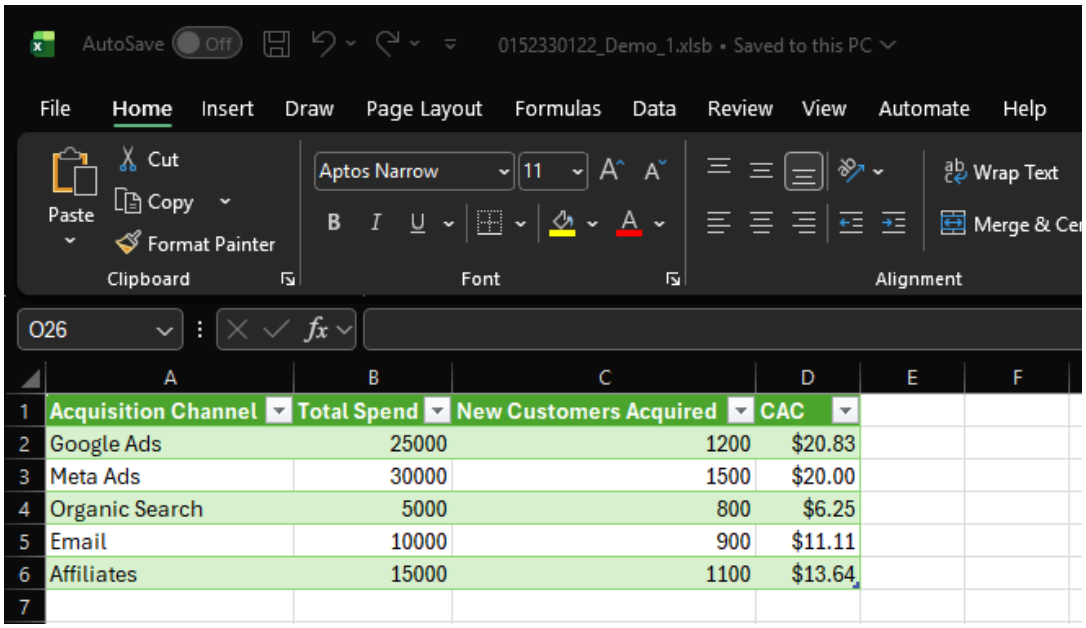
Step 2: Loading CSV and performing the ta Excel



	A	B	C	D	E	F	G	H	I	J
1	Transaction ID	Customer ID	Date	Product ID	Units Sold	Acquisition Channel	Unit Pri	Gross Revenu	Cohort	Channe
2	TXN0000001	CUST7065	2/12/2025	P043	1	Email	$64.70	$64.70	2025-02	$11.11
3	TXN0000002	CUST6820	6/26/2025	P004	5	Meta Ads	$63.57	$317.85	2025-06	$20.00
4	TXN0000003	CUST4432	12/6/2025	P007	1	Organic Search	$224.12	$224.12	2025-12	$6.25
5	TXN0000004	CUST5374	4/27/2025	P037	5	Meta Ads	$209.06	$1,045.30	2025-04	$20.00
6	TXN0000005	CUST2169	5/24/2025	P038	5	Google Ads	$158.44	$792.20	2025-05	$20.83
7	TXN0000006	CUST3803	4/23/2025	P044	1	Organic Search	$79.45	$79.45	2025-04	$6.25
8	TXN0000007	CUST9751	4/29/2025	P019	2	Email	$204.26	$408.52	2025-04	$11.11
9	TXN0000008	CUST5010	7/26/2025	P011	2	Affiliates	$62.47	$124.94	2025-07	$13.64
10	TXN0000009	CUST3677	7/17/2025	P022	1	Affiliates	$177.55	$177.55	2025-07	$13.64
11	TXN0000010	CUST8573	9/1/2025	P017	4	Email	$62.91	$251.64	2025-09	$11.11
12	TXN0000011	CUST7216	2/23/2025	P040	5	Meta Ads	$148.56	$742.80	2025-02	$20.00
13	TXN0000012	CUST5422	12/10/2025	P032	5	Email	$185.14	$925.70	2025-12	$11.11
14	TXN0000013	CUST4598	3/31/2025	P015	3	Meta Ads	$165.97	$497.91	2025-03	$20.00
15	TXN0000014	CUST6313	10/10/2025	P018	4	Organic Search	$151.42	$605.68	2025-10	$6.25
16	TXN0000015	CUST1916	9/29/2025	P003	1	Meta Ads	$76.01	$76.01	2025-09	$20.00
17	TXN0000016	CUST4752	2/10/2025	P046	2	Email	$65.87	$131.74	2025-02	$11.11
18	TXN0000017	CUST1525	1/25/2025	P021	2	Meta Ads	$203.40	$406.80	2025-01	$20.00
19	TXN0000018	CUST6168	12/17/2025	P022	2	Email	$177.55	$355.10	2025-12	$11.11
20	TXN0000019	CUST7572	8/13/2025	P039	4	Organic Search	$216.81	$867.24	2025-08	$6.25
21	TXN0000020	CUST5386	10/21/2025	P037	1	Google Ads	$209.06	$209.06	2025-10	$20.83
22	TXN0000021	CUST2084	2/7/2025	P022	5	Google Ads	$177.55	$887.75	2025-02	$20.83
23	TXN0000022	CUST4456	1/10/2025	P044	3	Email	$79.45	$238.35	2025-01	$11.11
24	TXN0000023	CUST6155	9/25/2025	P035	1	Affiliates	$100.85	$100.85	2025-09	$13.64
25	TXN0000024	CUST4483	5/10/2025	P013	5	Affiliates	$16.37	$81.85	2025-05	$13.64
26	TXN0000025	CUST9179	9/19/2025	P047	4	Affiliates	$34.24	$136.96	2025-09	$13.64
27	TXN0000026	CUST7482	3/11/2025	P020	3	Meta Ads	$11.56	$34.68	2025-03	$20.00
28	TXN0000027	CUST8517	6/4/2025	P008	5	Email	$30.87	$154.35	2025-06	$11.11
29	TXN0000028	CUST3340	2/25/2025	P022	4	Organic Search	$177.55	$710.20	2025-02	$6.25
30	TXN0000029	CUST5339	9/12/2025	P034	5	Google Ads	$243.55	$1,217.75	2025-09	$20.83
31	TXN0000030	CUST3287	10/17/2025	P018	2	Google Ads	$151.42	$302.84	2025-10	$20.83
32	TXN0000031	CUST5040	5/25/2025	P009	3	Meta Ads	$111.26	$333.78	2025-05	$20.00
33	TXN0000032	CUST9830	7/12/2025	P007	3	Organic Search	$224.12	$672.36	2025-07	$6.25
34	TXN0000033	CUST5304	12/16/2025	P004	4	Email	$63.57	$254.28	2025-12	$11.11
35	TXN0000034	CUST8019	7/11/2025	P001	5	Email	$163.46	$817.30	2025-07	$11.11
36	TXN0000035	CUST7543	4/5/2025	P037	5	Meta Ads	$209.06	$1,045.30	2025-04	$20.00

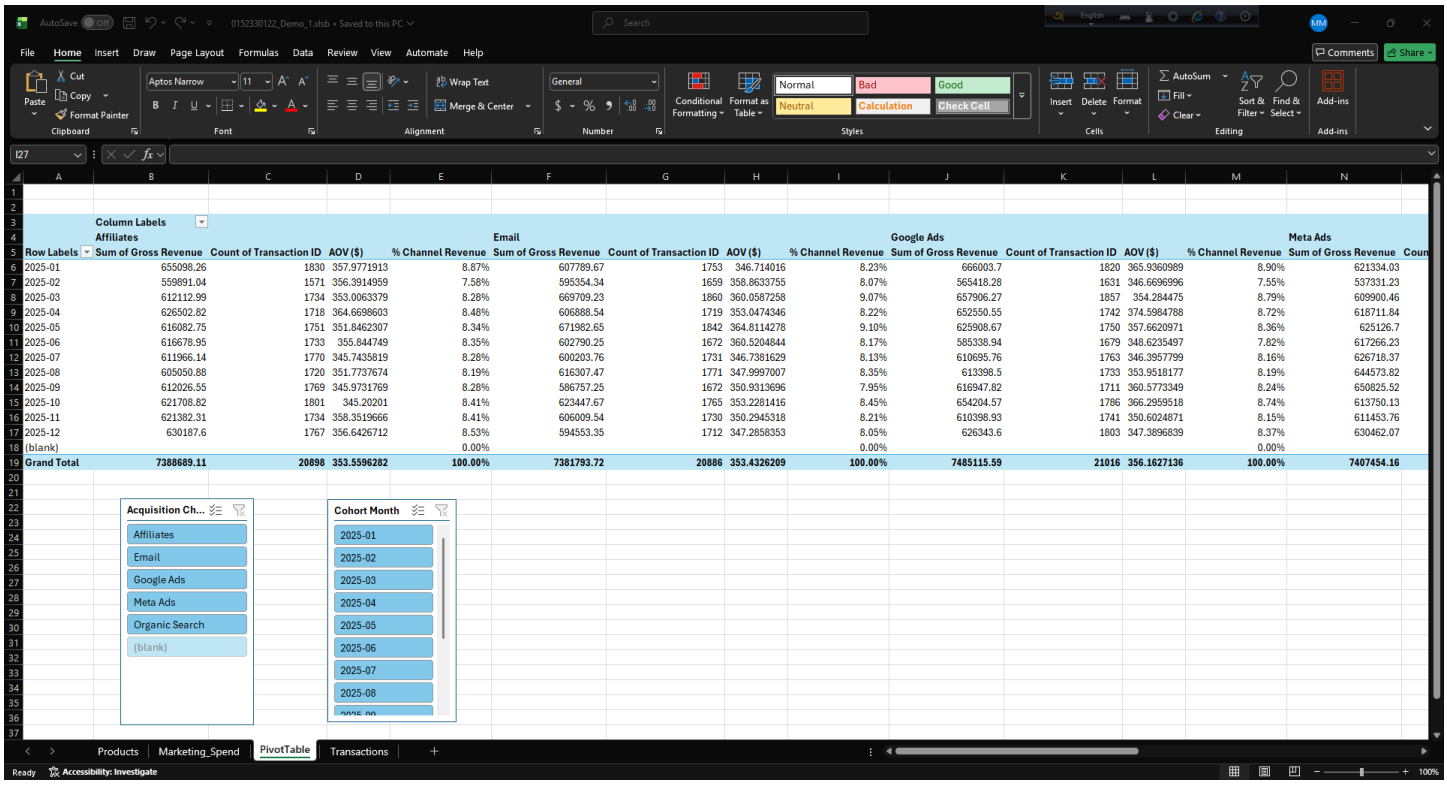


Product ID	Product Name	Unit Price
P033	Product 33	138.69
P034	Product 34	243.55
P035	Product 35	100.85
P036	Product 36	142.49
P037	Product 37	209.06
P038	Product 38	158.44
P039	Product 39	216.81
P040	Product 40	148.56
P041	Product 41	179.1
P042	Product 42	21
P043	Product 43	64.7
P044	Product 44	79.45
P045	Product 45	29.15
P046	Product 46	65.87
P047	Product 47	34.24
P048	Product 48	76.71
P049	Product 49	162.56
P050	Product 50	97.56



Acquisition Channel	Total Spend	New Customers Acquired	CAC
Google Ads	25000	1200	$20.83
Meta Ads	30000	1500	$20.00
Organic Search	5000	800	$6.25
Email	10000	900	$11.11
Affiliates	15000	1100	$13.64

Step 3: Final PivotTable



Column Labels	Affiliates				Email				Google Ads				Meta Ads	
Row Labels	Sum of Gross Revenue	Count of Transaction ID	AOV ($)	% Channel Revenue	Sum of Gross Revenue	Count of Transaction ID	AOV ($)	% Channel Revenue	Sum of Gross Revenue	Count of Transaction ID	AOV ($)	% Channel Revenue	Sum of Gross Revenue	Coun
2025-01	655098.26	1830	357.9771913	8.87%	607789.67	1753	346.714016	8.23%	666003.7	1820	365.9360989	8.90%	621334.03	
2025-02	559891.04	1571	356.3914959	7.58%	595354.34	1659	358.8633755	8.07%	565418.28	1631	346.6696996	7.55%	537331.23	
2025-03	612112.99	1734	353.0063379	8.28%	669709.23	1860	360.0587258	9.07%	657906.27	1857	354.284475	8.79%	609900.46	
2025-04	626502.82	1718	364.6698603	8.48%	606888.54	1719	353.0474346	8.22%	652550.55	1742	374.5984788	8.72%	618711.84	
2025-05	616082.75	1751	351.8462307	8.34%	671982.65	1842	364.8114278	9.10%	625908.67	1750	357.6620971	8.36%	625126.7	
2025-06	616678.95	1733	355.844749	8.35%	602790.25	1672	360.5204844	8.17%	585338.94	1679	348.6235497	7.82%	617266.23	
2025-07	611966.14	1770	345.7435819	8.28%	600203.76	1731	346.7381629	8.13%	610695.76	1763	346.3957799	8.16%	626718.37	
2025-08	605050.88	1720	351.7737674	8.19%	616307.47	1771	347.9997007	8.35%	613398.5	1733	353.9518177	8.19%	644573.82	
2025-09	612026.55	1769	345.9731769	8.28%	586757.25	1672	350.9313696	7.95%	616947.82	1711	360.5773349	8.24%	650825.52	
2025-10	621708.82	1801	345.20201	8.41%	623447.67	1765	353.2281416	8.45%	654204.57	1786	366.2959518	8.74%	613750.13	
2025-11	621382.31	1734	358.3519666	8.41%	606009.54	1730	350.2945318	8.21%	610398.93	1741	350.6024871	8.15%	611453.76	
2025-12	630187.6	1767	356.6426712	8.53%	594553.35	1712	347.2858353	8.05%	626343.6	1803	347.3896839	8.37%	630462.07	
(blank)				0.00%				0.00%				0.00%		
Grand Total	**7388689.11**	**20898**	**353.5596282**	**100.00%**	**7381793.72**	**20886**	**353.4326209**	**100.00%**	**7485115.59**	**21016**	**356.1627136**	**100.00%**	**7407454.16**	

Acquisition Ch...
- Affiliates
- Email
- Google Ads
- Meta Ads
- Organic Search
- (blank)

Cohort Month
- 2025-01
- 2025-02
- 2025-03
- 2025-04
- 2025-05
- 2025-06
- 2025-07
- 2025-08

Products | Marketing_Spend | PivotTable | Transactions